Jaeho Jeon 전재호 · 田宰昊

Ph.D. Student, Department of Electrical Engineering and Computer Science, DGIST

Privacy and Applied Cryptography Lab (PACL) · Advisor: Prof. Young-Sik Kim

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RESEARCH INTERESTS

Side-channel analysis · Hardware security · Post-quantum cryptography implementation · Cryptographic engineering · Computer architecture · Coding theory

My research focuses on **side-channel analysis**. My goal is to become a uniquely recognized expert in side-channel analysis, and to gradually expand into computer architecture, quantum computing, networking, and coding theory.

EDUCATION

2026.02 – present	**Ph.D. student**, Electrical Engineering and Computer Science, DGIST
2024.02 – 2026.02	**M.S.**, Electrical Engineering and Computer Science, DGIST
2017.02 – 2024.02	**B.S.**, School of Undergraduate Studies, DGIST
2014.03 – 2017.02	Cheonan Upsung High School, Cheonan

PUBLICATIONS

RB-LDC: Redundancy-Balanced Latent Coding for Robust Diffusion. Hyunseok Jeong, **Jaeho Jeon**, Young-Sik Kim. *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted.

Codeword Masking Can Be Harmful Under Replay Attacks on HQC. **Jaeho Jeon**, Yongseong Park, Jaeyeon Lee, Suseong Lee, Donghyen Kim, Young-Sik Kim. *International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2026)*, accepted. Cryptology ePrint Archive 2026/071.

HONORS & AWARDS

2026.09	**Grand Prize, 4th KICS HFR Paper Award** (제4회 KICS HFR 논문대상), Korean Institute of Communications and Information Sciences. Yongseong Park, Hyunseok Jeong, Jaeho Jeon. Total prize $10,000.
2026.06	**GRAVITY 2026 Startup Competition**, passed the DGIST preliminary round with a hardware-hacking (side-channel) certification service for cryptographic products. Prize $10,000.

TEACHING & MENTORING

2026.06 – 2026.09	**Teaching Assistant**, Pre-URP, DGIST. Mentored a student team on *"Reading the Current to Break Ciphers: Side-Channel Analysis on RSA/AES"*.
2025.03 – 2025.12	**Teaching Assistant**, UGRP (Undergraduate Group Research Program), DGIST. Supported the team that built an optimized PQC implementation on ARM Cortex-M4. The team won two Grand Prizes: PQC Migration Challenge (NIA, 2025.11) and Undergraduate Best Paper Award, 2025 KICS Fall Conference.

ACADEMIC & STUDENT SERVICES

2026.07	**Staff**, DGIST AI & Crypto Summer Camp 2026 — organized the camp, hosted the rump session
2026.04 – 2026.09	**President**, DGIST Graduate Student Council — representing graduate students of all departments
2026.03 – present	**Graduate Student Representative**, Dept. of EECS, DGIST
2024.10	**Treasurer**, 2024 EECS & AI DGIST Student Conference & Happy Hour

ASIACRYPT 2026 · STUDENT STIPEND APPLICATION

Student Statement

Jaeho Jeon · Ph.D. student, Department of Electrical Engineering and Computer Science, DGIST (Daegu, South Korea) · Privacy and Applied Cryptography Lab (PACL), advised by Prof. Young-Sik Kim · dgwogh@dgist.ac.kr

Presenting: **"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"** (first author), accepted to Asiacrypt 2026

Summary of research interests

My research focuses on **side-channel analysis** and the implementation security of cryptographic systems. I study how secret information leaks from devices through physical channels such as power consumption, and how countermeasures that look sound on paper can fail, or even backfire, when they meet a concrete attacker model. My recent work centers on **post-quantum cryptography (PQC)**, in particular code-based schemes: our Asiacrypt 2026 paper shows that codeword masking, a widely used countermeasure in HQC implementations, can actually be harmful under replay attacks, and analyzes the conditions under which it reduces rather than improves security. I have also worked on PQC implementations on constrained devices (ARM Cortex-M4), and on coding-theoretic tools for robustness, which led to a paper accepted to NeurIPS 2026.

My long-term goal is to become a recognized expert in side-channel analysis and to grow outward from it into computer architecture, networking, and coding theory, so that the security of a cryptographic system can be assessed end to end, from the algorithm down to the silicon.

How I will benefit from participating in Asiacrypt

Asiacrypt 2026 will be my first time presenting at an IACR flagship conference. I am a first-year Ph.D. student, and this paper is the first work I have led from the initial observation to the final analysis. Presenting it in person to the researchers who design, implement, and attack code-based schemes is the most direct way to test our claims, to learn which assumptions the community finds convincing, and to understand which follow-up questions matter most. Since HQC was selected by NIST for standardization, the question of which countermeasures are safe to deploy has become urgent, and I expect the discussions around our result to shape the next steps of my thesis.

Beyond my own talk, Asiacrypt is where the PQC, implementation-security, and side-channel communities meet. My lab is relatively small and geographically distant from most of these groups, so a few days of conversations at the conference are worth months of reading papers alone. I also mentor undergraduate research teams at DGIST and help organize our lab's cryptography summer camp, so what I learn at Asiacrypt will be passed on to students deciding whether to pursue cryptography.

Financial need

As a student presenter I have applied for the registration fee waiver provided by the Cryptography Research Fund. The remaining cost of attending, round-trip travel from Daegu to Hong Kong and accommodation for the conference week, is not covered. My lab's travel budget for this year is limited, and my advisor can contribute at most [amount in USD] toward this trip (see the accompanying letter). I estimate the total cost at approximately [estimated total in USD] ([flight] for airfare and [hotel] for lodging), and I am requesting a stipend to help cover the difference. I have no other external funding source for conference travel, and without support attending in person would be very difficult.

Thank you for considering my application.

Jaeho Jeon · [date]

DGIST · Privacy and Applied Cryptography Lab (PACL)
Department of Electrical Engineering and Computer Science, Daegu Gyeongbuk Institute of Science and Technology, Daegu, South Korea

[Date]

To the Asiacrypt 2026 Student Stipend Committee

Re: Letter of support and justification of financial need for Jaeho Jeon

I am writing to support the student stipend application of my Ph.D. student, **Jaeho Jeon**, who will present the paper *"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"* at Asiacrypt 2026 as its first author.

Jaeho joined my laboratory as a master's student in 2024 and began his Ph.D. in February 2026. He works on side-channel analysis and the implementation security of post-quantum cryptography. The Asiacrypt paper is his own initiative: he noticed that a standard countermeasure for HQC behaves unexpectedly under replay attacks, developed the analysis, and carried the work through to acceptance. He is also a co-author of a paper accepted to NeurIPS 2026 and received the Grand Prize of the 4th KICS HFR Paper Award in 2026. In addition to research, he mentors undergraduate research teams, one of which won two national-level awards for a PQC implementation on ARM Cortex-M4 under his guidance, and he organized our lab's AI & Crypto Summer Camp this year. He is exactly the kind of early-career researcher who benefits most from presenting at, and meeting the community of, an IACR conference.

Justification of financial need. Our laboratory's conference travel budget for 2026 is limited and has been largely allocated to other presentations earlier in the year. For Jaeho's attendance at Asiacrypt 2026 in Hong Kong, the laboratory and the institution can provide at most [maximum amount in USD] toward his travel and accommodation; [we can / cannot] provide additional funds beyond this amount. Jaeho has applied for the student presenter registration fee waiver, so the remaining need is for airfare and lodging, which we estimate at [estimated total in USD]. Without a stipend, the shortfall of approximately [shortfall in USD] would fall on the student himself, which is not reasonable for a first-year Ph.D. student on a standard stipend.

I therefore ask the committee to consider Jaeho for a travel stipend. I am confident that he will make full use of the opportunity, and that his participation will benefit both his development and the discussion of countermeasures for code-based PQC at the conference. Please do not hesitate to contact me if any further information is needed.

Sincerely,

Young-Sik Kim, Professor
Department of Electrical Engineering and Computer Science, DGIST
Privacy and Applied Cryptography Lab (PACL)
[advisor email] · [phone]